Medhansh Garg

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Education

University of Illinois, Urbana-Champaign Aug. 2023 – May 2027 (Expected)
- B.S. in Computer Engineering | Relevant Coursework: Data Structures & Algorithms, Computer Systems Engineering, Computer Architecture, Operating Systems, Databases

Skills

- **Languages**: TypeScript, Python, Java, C, C++, SQL, Kotlin, Go
- **Frameworks & Web**: React, Next.js, Node.js, FastAPI, Flask, Spring Boot, Prisma, SQLAlchemy, REST APIs
- **Data & Cloud**: PostgreSQL, SQLite, Supabase, Docker, Kubernetes, AWS, Azure, GitHub Actions CI/CD, Git, Linux
- **Systems**: RISC-V, QEMU, WebAssembly, Emscripten, Android HALs
- **Security**: Burp Suite, Frida, JADX, OAuth2/CILogon SSO, Row-Level Security, RBAC, Tailscale, nginx

Experience

Care Health Insurance, Gurugram, India | *Cybersecurity Intern* Jun. 2025 – Aug. 2025
- Ran external recon that uncovered **60+ unauthenticated exposed services**, **30** of which were pulled internal after my findings
- Reverse-engineered **WAF request-signing** into a **Burp Suite** replay PoC, and bypassed **SSL pinning** and root detection (**Frida**, **JADX**) to unlock dynamic analysis of the customer mobile app
- Assessed the customer portal and Android app with **Nmap**, **Wireshark**, **Burp Suite**, **Frida**, **JADX**

CS 124 Honors, Urbana, IL | *Lead Web Developer* Aug. 2025 – Present
- Lead development of the CS 124 platform (**Next.js**, **Supabase**), hosted on **Vercel** with **GitHub Actions** CI deploying directly to it, serving **300+ students** and **30+ staff** each semester. Built **CILogon** SSO and a 6-role least-privilege system with temporary role-view access, no permanent escalation. Lead an **11-person** dev team
- Built a **TOTP** attendance system with **30-second** rotating one-time codes, so only students in the room can check in
- Routed all public pages through a server-side API layer so no DB keys reach the client. Built a **246-test Vitest/Playwright** suite (6 unit, 167 API/DB integration, 73 browser E2E), with E2E gating dev-to-main promotions in **GitHub Actions** CI

Hopcharge, Gurugram, India | *Software Engineering Intern* Jun. 2026 – Aug. 2026
- Built **VOLT.CV**, a **FastAPI** HR dashboard automating hiring (resume parsing with 3-tier OCR fallback, Gmail reply classification, bulk interview scheduling), secured with database-level **Row-Level Security** tied to Google identities in **Supabase/PostgreSQL**
- Built the **Ad Engine** (**Next.js**, **React**, **Prisma/Postgres**) publishing ad creatives to **Meta** and **YouTube**; scaled output from 2–3 ads/week to **3/day**, about **8**×

IoT++, Remote | *Software Engineering Intern* Jun. 2024 – Aug. 2024
- Built a real-time fire/smoke detection pipeline with **YOLOv8**, **62%** better accuracy relative to the base YOLO model; shipped **NPU** inference on **12 Orange Pi units** in a factory pilot

Projects

ECE 391 RISC-V Kernel | *C, RISC-V, QEMU* github.com/HackOverflow404/ece391-kernel-demo
- With a team, wrote a **RISC-V64 kernel** from scratch: user/kernel syscalls, page-table virtual memory, scheduler with kernel threads, **VirtIO** console and block drivers, **KTFS** filesystem, and a Unix-style shell with multi-stage pipelines
- Compiled **TinyEMU** to **WebAssembly** (Emscripten) so the kernel boots live in the browser, with an **xterm.js** shell and filesystem mounted over HTTP: ece-391-kernel-demo.vercel.app

Echo Show 5: LineageOS & Camera Pipeline | *C++, Kotlin, HALs* github.com/HackOverflow404/EchoCameraStreamer
- Ported **LineageOS 18.1** via the **amonet bootrom exploit**; fixed a **MediaTek camera ABI/vtable mismatch** and a defconfig bug
- Built a **Kotlin 720p H.264** webcam app streaming over **Tailscale**, self-recovering from camera, encoder and network failures; a full reboot comes back live in **55s**

Mainframe & SECUR-T | *Raspberry Pi, Tailscale, go2rtc, MQTT* Portfolio: Mainframe entry
- Built laptop face unlock: Pi returns **Ed25519**-signed verdicts bound to host, one-time nonce and 30s window, via **MQTT** into **PAM**
- Built **SECUR-T**, a Raspberry Pi hub that fans one live Echo Show camera stream (24 fps) out to **go2rtc**, **Home Assistant** and a recorder; recorder writes 10-minute segments to a NAS with no re-encoding, **25GB** rolling buffer (about 18h), about **7%** Pi CPU
- Operate a **Pi 4** homelab since May 2025 (17 months, about **97%** uptime), reachable only over a 3-device **Tailscale** mesh with no router ports opened: **11** self-hosted services (**Home Assistant** in **Docker**, **MQTT** broker, **Samba** NAS, **systemd** services), about 1TB of a 2TB NAS in use

Certifications & Publications

NYUx MicroBachelors in Cybersecurity Fundamentals | *Certification* Jun. 2022 – Apr. 2023
- Completed a 9-course program (**A grade**) covering authentication and access control, network security, and penetration testing

Biometric Authentication and Cybersecurity in the Digital Age | *Published review paper* medcrave.com
- Compared biometrics with passwords and tokens for online banking; concluded layering with one-time passwords is strongest

Leadership & Involvement

Illinois MicroTech, Urbana, IL | *External Vice President* | *EOH Project Lead* Oct. 2025 – Present
- Led a **15-person** team delivering the club's Engineering Open House 2026 showcase to **20,000+ attendees** on a **$250** budget
- Manage the club's marketing chairs, EOH leads, and outreach committee, alongside independent outreach initiatives
- HackerFab: engineering a **3-axis stepper-driven stage** with **sub-nm positioning** for wafer photolithography